Volume Vi Reference

Contents

Introduction to Algebraic Structures

0.1 What Is a Mathematical Structure?

0.1.1 Algebraic Structures

Definition 0.1.1 (Algebraic Structure)**.** An **algebraic structure** of signature $\mathcal{L}$ is a pair

$$\mathcal{A} = (A, I)$$

where A is a nonempty carrier set and I interprets each symbol of $\mathcal{L}$ on A: each n-ary function symbol as an operation $A^n \to A$, each n-ary relation symbol as a subset of A^n, and each constant symbol as an element of A. The structure is algebraic when $\mathcal{L}$ has no relation symbols beyond equality.

Definition 0.1.2 (First-Order Signature)**.** A **first-order signature**

$$\mathcal{L} = (\mathrm{Const}_{\mathcal{L}}, \mathrm{Func}_{\mathcal{L}}, \mathrm{Rel}_{\mathcal{L}}, \mathrm{arity})$$

lists constant symbols, function symbols, and relation symbols, with each function or relation symbol carrying a declared arity.

Definition 0.1.3 ($\mathcal{L}$-Structure)**.** Given a signature $\mathcal{L}$, an **$\mathcal{L}$-structure** is a nonempty carrier set A together with an interpretation of every symbol of $\mathcal{L}$ on A.

0.1.2 Laws of Composition

Definition 0.1.4 (Binary Operation / Law of Composition)**.** Let A be a set. A **binary operation** on A (Bourbaki: an internal law of composition) is a function

$$\star : A \times A \to A.$$

The image of (a, b) is written $a \star b$.

Definition 0.1.5 (n-ary Operation)**.** An **n-ary operation** on A is a function $f : A^n \to A$, with $n \geq 0$. It is unary if $n = 1$, binary if $n = 2$, and nullary if $n = 0$.

Definition 0.1.6 (Unary Operation)**.** A **unary operation** on A is a function $f : A \to A$.

Definition 0.1.7 (Nullary Operation / Constant)**.** A **nullary operation** on A is the selection of a distinguished element $c \in A$; equivalently, it is an operation $A^0 \to A$.

0.1.3 Distinguished Elements

Definition 0.1.8 (Left Identity)**.** An element $e_L \in A$ is a **left identity** for $\star$ if $e_L \star a = a$ for all $a \in A$.

Definition 0.1.9 (Right Identity)**.** An element $e_R \in A$ is a **right identity** for $\star$ if $a \star e_R = a$ for all $a \in A$.

Definition 0.1.10 (Identity / Neutral Element)**.** An element $e \in A$ is a **two-sided identity** (or neutral element) for $\star$ if it is both a left and a right identity:

$$e \star a = a \star e = a \quad \text{for all } a \in A.$$

Proposition 0.1.11 (Identity Collapse and Uniqueness)**.** *If $\star$ has a left identity e_L and a right identity e_R, then $e_L = e_R$. Consequently a two-sided identity, if it exists, is unique.* *Go to proof.*

Definition 0.1.12 (Left Absorbing Element)**.** An element $z_L \in A$ is **left absorbing** for $\star$ if $z_L \star a = z_L$ for all $a \in A$.

Definition 0.1.13 (Right Absorbing Element)**.** An element $z_R \in A$ is **right absorbing** for $\star$ if $a \star z_R = z_R$ for all $a \in A$.

Definition 0.1.14 (Absorbing Element)**.** An element $z \in A$ is **absorbing** (a zero element) for $\star$ if it is both left absorbing and right absorbing.

Proposition 0.1.15 (Uniqueness of the Absorbing Element)**.** *A two-sided absorbing element, if it exists, is unique.* *Go to proof.*

0.1.4 Inverses

Definition 0.1.16 (Left Inverse)**.** Let $\star$ have identity e, and let $a \in A$. An element $b \in A$ is a **left inverse** of a if $b \star a = e$.

Definition 0.1.17 (Right Inverse)**.** Let $\star$ have identity e, and let $a \in A$. An element $c \in A$ is a **right inverse** of a if $a \star c = e$.

Definition 0.1.18 (Inverse)**.** Let $\star$ have identity e, and let $a \in A$. An element $a^{-1} \in A$ is an **inverse** of a if

$$a \star a^{-1} = a^{-1} \star a = e.$$

Definition 0.1.19 (Invertible Element)**.** An element a is **invertible** (a unit) if it has a two-sided inverse.

Proposition 0.1.20 (Inverse Collapse and Uniqueness in a Monoid)**.** *If $\star$ is associative with identity e, and a has a left inverse b and a right inverse c, then $b = c$. Hence the inverse of an invertible element is unique.* *Go to proof.*

Definition 0.1.21 (Additive Inverse)**.** When $\star = +$ with identity 0, the inverse of a is written $-a$ and is called the **additive inverse** or negative.

Definition 0.1.22 (Multiplicative Inverse)**.** When $\star = \cdot$ with identity 1, the inverse of a is written a^{-1} and is called the **multiplicative inverse**.

Proposition 0.1.23 (Units of a Monoid Form a Group)**.** *In a monoid $(A, \star, e)$, the set $A^\times$ of invertible elements is closed under $\star$ and forms a group.* *Go to proof.*

0.1.5 Properties of Laws

Definition 0.1.24 (Associativity)**.** A binary operation $\star$ on A is **associative** if

$$(a \star b) \star c = a \star (b \star c) \quad \text{for all } a, b, c \in A.$$

Theorem 0.1.25 (General Associativity)**.** *If $\star$ is associative, then any finite product $a_1 \star a_2 \star \cdots \star a_n$ has a value independent of parenthesization.* *Go to proof.*

Definition 0.1.26 (Powers / Iterated Operation)**.** For associative $\star$ with identity e, define $a^0 := e$ and $a^{n+1} := a^n \star a$. In additive notation, define $0a := 0$ and $(n+1)a := na + a$.

Definition 0.1.27 (Commutativity)**.** A binary operation $\star$ on A is **commutative** if $a \star b = b \star a$ for all $a, b \in A$.

Definition 0.1.28 (Left Distributivity)**.** For laws $+$ and $\cdot$ on A, multiplication is **left distributive** over addition if

$$a \cdot (b + c) = a \cdot b + a \cdot c$$

for all $a, b, c \in A$.

Definition 0.1.29 (Right Distributivity)**.** For laws $+$ and $\cdot$ on A, multiplication is **right distributive** over addition if

$$(a + b) \cdot c = a \cdot c + b \cdot c$$

for all $a, b, c \in A$.

Definition 0.1.30 (Distributivity)**.** For laws $+$ and $\cdot$ on A, multiplication is **distributive** over addition if it is both left distributive and right distributive.

Definition 0.1.31 (Idempotent Operation)**.** A binary operation $\star$ on A is **idempotent** if $a \star a = a$ for all $a \in A$.

0.1.6 Regularity Properties

Definition 0.1.32 (Left Cancellable Element)**.** An element $a \in A$ is **left cancellable** if

$$a \star x = a \star y \Longrightarrow x = y$$

for all $x, y \in A$.

Definition 0.1.33 (Right Cancellable Element)**.** An element $a \in A$ is **right cancellable** if

$$x \star a = y \star a \Longrightarrow x = y$$

for all $x, y \in A$.

Definition 0.1.34 (Cancellable Element)**.** An element $a \in A$ is **cancellable** (regular) if it is both left cancellable and right cancellable.

Proposition 0.1.35 (Invertible Implies Cancellable)**.** *In a monoid, every invertible element is cancellable.* *Go to proof.*

Definition 0.1.36 (Zero Divisor)**.** In a ring R, a nonzero element a is a **zero divisor** if there is a nonzero b with $a \cdot b = 0$ or $b \cdot a = 0$.

0.1.7 One Internal Law

Definition 0.1.37 (Magma)**.** A **magma** is a pair $(A, \star)$ where $A \neq \varnothing$ and $\star$ is a binary operation on A.

Definition 0.1.38 (Semigroup)**.** A **semigroup** is a magma whose operation is associative.

Definition 0.1.39 (Monoid)**.** A **monoid** is a semigroup with an identity element.

Definition 0.1.40 (Group)**.** A **group** is a monoid in which every element has an inverse.

Definition 0.1.41 (Abelian Group)**.** An **abelian group** is a group whose operation is commutative.

Definition 0.1.42 (Order of a Group)**.** The **order** of a group G, denoted $|G|$, is the cardinality of its carrier. The group is finite if $|G|$ is finite.

Proposition 0.1.43 (Uniqueness of the Identity in a Group)**.** *The identity of a group is unique.* *Go to proof.*

Proposition 0.1.44 (Uniqueness of Inverses in a Group)**.** *Each element of a group has a unique inverse.* *Go to proof.*

Proposition 0.1.45 (Cancellation Laws in a Group)**.** *In a group, $ab = ac$ implies $b = c$, and $ba = ca$ implies $b = c$.* *Go to proof.*

Proposition 0.1.46 (Socks–Shoes)**.** *In a group, $(ab)^{-1} = b^{-1}a^{-1}$.* *Go to proof.*

0.1.8 Two Internal Laws

Definition 0.1.47 (Rng)**.** A **rng** is a triple $(R, +, \cdot)$ where $(R, +)$ is an abelian group, $\cdot$ is associative, and $\cdot$ distributes over $+$. No multiplicative identity is required.

Definition 0.1.48 (Ring)**.** A **ring** is a rng with a multiplicative identity 1; equivalently, $(R, \cdot, 1)$ is a monoid. Thus the axioms are: additive abelian group, multiplicative associativity, distributivity, and unity.

Definition 0.1.49 (Commutative Ring)**.** A **commutative ring** is a ring whose multiplication is commutative.

Definition 0.1.50 (Integral Domain)**.** An **integral domain** is a commutative ring with $0 \neq 1$ and no zero divisors.

Definition 0.1.51 (Division Ring / Skew Field)**.** A **division ring** (or skew field) is a ring with $0 \neq 1$ in which every nonzero element is invertible under multiplication. Multiplication is not required to be commutative.

Definition 0.1.52 (Field)**.** A **field** is a commutative division ring; equivalently, it is a commutative ring with $0 \neq 1$ in which $(F \setminus \{0\}, \cdot)$ is a group.

Proposition 0.1.53 (Multiplication by Zero)**.** *In a ring, $a \cdot 0 = 0 \cdot a = 0$ for all a.* *Go to proof.*

Proposition 0.1.54 (Multiplication by Additive Inverse)**.** *In a ring,*

$$a \cdot (-b) = -(a \cdot b) = (-a) \cdot b.$$

In a ring, $(-1) \cdot a = -a$. *Go to proof.*

Proposition 0.1.55 (Cancellation in Integral Domains)**.** *In an integral domain, if $a \neq 0$ and $ab = ac$, then $b = c$.* *Go to proof.*

Proposition 0.1.56 (Every Field is an Integral Domain)**.** *Every field is an integral domain.* *Go to proof.*

Proposition 0.1.57 (Finite Integral Domain is a Field)**.** *Every finite integral domain is a field.* *Go to proof.*

0.1.9 The Structural Lattice

0.2 Relations

0.3 Operations

0.4 Laws of Operations

0.5 Functions

0.6 Distinguished Elements

0.7 Algebraic Structures

0.8 Groups

0.9 Rings

0.10 Fields

0.11 Number Systems as Structures

0.12 Number Systems as Models

Proofs

Capstone

Abstract Algebra

0.13 Induction

0.13.1 Induction

Definitions and Theorems

0.14 Integers

0.14.1 Integers

Basic Definitions and Theorems

0.15 Modular Arithmetic

0.15.1 Modular Arithmetic

Basic Definitions and Theorems

Proofs

Capstone

Algebraic Geometry

0.16 Algebraic Geometry

0.16.1 Preliminary Definitions

Definition 0.16.1 (Ring)**.** A set R with two binary operations $+ : R \times R \to R$ and $\cdot : R \times R \to R$ is a *ring* if:

1. **Additive structure:** $(R, +)$ is an abelian group.
2. **Associativity of multiplication:**
$$(a \cdot b) \cdot c = a \cdot (b \cdot c) \quad \text{for all } a, b, c \in R.$$
3. **Multiplicative identity:** There exists $1 \in R$ such that
$$1 \cdot a = a \cdot 1 = a \quad \text{for all } a \in R.$$
4. **Distributivity:**
$$a \cdot (b + c) = a \cdot b + a \cdot c \quad \text{and} \quad (a + b) \cdot c = a \cdot c + b \cdot c \quad \text{for all } a, b, c \in R.$$

Definition 0.16.2 (Commutative Ring)**.** A ring R is *commutative* if
$$a \cdot b = b \cdot a \quad \text{for all } a, b \in R.$$

Polynomial Rings

Definition 0.16.3 (Polynomial)**.** Let R be a commutative ring. A *polynomial* in one variable over R is a formal expression
$$f = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_1 x + a_0,$$
where $n \in \mathbb{N}$, the *coefficients* $a_0, a_1, \ldots, a_n \in R$, and x is a formal symbol called an *indeterminate*.

Definition 0.16.4 (Degree)**.** Let $f = a_n x^n + \cdots + a_0$ be a polynomial over R. If $a_n \neq 0$, then the *degree* of f is
$$\deg(f) := n.$$
The coefficient a_n is called the *leading coefficient* of f. A polynomial with leading coefficient 1 is called *monic*.

Definition 0.16.5 (Polynomial Ring)**.** Let R be a commutative ring. The *polynomial ring* over R in one indeterminate x, denoted $R[x]$, is the set of all polynomials in x with coefficients in R, equipped with addition and multiplication defined by:
$$\left(\sum_i a_i x^i\right) + \left(\sum_i b_i x^i\right) := \sum_i (a_i + b_i) x^i,$$
$$\left(\sum_i a_i x^i\right) \cdot \left(\sum_j b_j x^j\right) := \sum_k \left(\sum_{i+j=k} a_i b_j\right) x^k.$$

Polynomial Rings in Several Variables

Definition 0.16.6 (Polynomial Ring in n Variables)**.** Let R be a commutative ring and let $n \in \mathbb{N}$. The *polynomial ring in n variables* over R, denoted

$$R[x_1, x_2, \ldots, x_n],$$

is defined inductively by

$$R[x_1, \ldots, x_n] := R[x_1, \ldots, x_{n-1}][x_n].$$

Its elements are finite sums of the form

$$f = \sum_{\alpha} a_\alpha \, x^\alpha,$$

where the sum ranges over multi-indices $\alpha = (\alpha_1, \ldots, \alpha_n) \in \mathbb{N}^n$, the coefficients $a_\alpha \in R$, and $x^\alpha := x_1^{\alpha_1} \cdots x_n^{\alpha_n}$.

Definition 0.16.7 (Monomial)**.** A *monomial* in $R[x_1, \ldots, x_n]$ is a polynomial of the form

$$x^\alpha = x_1^{\alpha_1} \cdots x_n^{\alpha_n}$$

for some multi-index $\alpha \in \mathbb{N}^n$.

Definition 0.16.8 (Total Degree)**.** The *total degree* of the monomial x^α is

$$|\alpha| := \alpha_1 + \alpha_2 + \cdots + \alpha_n.$$

The degree of a polynomial $f \in R[x_1, \ldots, x_n]$ is the maximum total degree among all monomials with nonzero coefficient.

Ideals

Definition 0.16.9 (Ideal)**.** Let R be a commutative ring. A subset $I \subseteq R$ is an *ideal* of R if:

1. $0 \in I$,
2. $a + b \in I$ for all $a, b \in I$,
3. $r \cdot a \in I$ for all $r \in R$ and $a \in I$.

Definition 0.16.10 (Generated Ideal)**.** Let R be a commutative ring and let $f_1, \ldots, f_m \in R$. The *ideal generated by $f_1, \ldots, f_m$* is

$$\langle f_1, \ldots, f_m \rangle := \left\{ \sum_{i=1}^{m} r_i f_i : r_i \in R \right\}.$$

An ideal of this form is called *finitely generated*.

Definition 0.16.11 (Finitely Generated Ideal)**.** An ideal $I \subseteq R$ is *finitely generated* if there exist $f_1, \ldots, f_m \in R$ such that $I = \langle f_1, \ldots, f_m \rangle$.

Definition 0.16.12 (Noetherian Ring)**.** A commutative ring R is *Noetherian* if every ideal of R is finitely generated.

Theorem 0.16.13 (Hilbert Basis Theorem)**.** *If R is a Noetherian ring, then the polynomial ring $R[x]$ is also Noetherian.*
In particular, $k[x_1, \ldots, x_n]$ is Noetherian for any field k. *Go to proof.*

Quotient Rings

Definition 0.16.14 (Quotient Ring)**.** Let R be a commutative ring and let $I \subseteq R$ be an ideal. The *quotient ring* R/I is the set of cosets

$$R/I := \{a + I : a \in R\},$$

equipped with addition and multiplication defined by

$$(a + I) + (b + I) := (a + b) + I,$$

$$(a + I) \cdot (b + I) := (a \cdot b) + I.$$

Proofs

Capstone

Linear Algebra

0.17 Vector Space Preliminaries

Definition 0.17.1 (List)**.** Suppose n is a non-negative integer. A **list** of length n is an ordered collection of n elements.

Definition 0.17.2 ($\mathbf{F}^n$)**.** Let $\mathbf{F}$ denote $\mathbb{R}$ or $\mathbb{C}$, and let n be a positive integer. Then $\mathbf{F}^n$ is defined to be the set of all lists of length n of elements of $\mathbf{F}$:

$$\mathbf{F}^n = \{(x_1, \ldots, x_n) : x_1, \ldots, x_n \in \mathbf{F}\}.$$

The entries in a list in $\mathbf{F}^n$ are called the coordinates of the list.

Definition 0.17.3 (Addition in $\mathbf{F}^n$)**.** Let

$$x = (x_1, \ldots, x_n), \qquad y = (y_1, \ldots, y_n)$$

be vectors in $\mathbf{F}^n$. The sum of x and y is defined by

$$x + y = (x_1 + y_1, \ldots, x_n + y_n).$$

Theorem 0.17.4 (Addition Is Commutative in $\mathbf{F}^n$)**.** *For all $x, y \in \mathbf{F}^n$,*

$$x + y = y + x.$$

Go to proof.

Definition 0.17.5 (Zero Vector in $\mathbf{F}^n$)**.** The symbol 0 denotes the list of length n all of whose coordinates are 0:

$$0 = (0, \ldots, 0) \in \mathbf{F}^n.$$

Definition 0.17.6 (Additive Inverse in $\mathbf{F}^n$)**.** Let

$$x = (x_1, \ldots, x_n) \in \mathbf{F}^n.$$

The additive inverse of x is defined by

$$-x = (-x_1, \ldots, -x_n).$$

Definition 0.17.7 (Scalar Multiplication in $\mathbf{F}^n$)**.** Let

$$\lambda \in \mathbf{F} \quad \text{and} \quad x = (x_1, \ldots, x_n) \in \mathbf{F}^n.$$

The product of λ and x is defined by

$$\lambda x = (\lambda x_1, \ldots, \lambda x_n).$$

Definition 0.17.8 (Vector Space)**.** A **vector space** over $\mathbf{F}$ is a set V together with an addition on V and a scalar multiplication

$$\mathbf{F} \times V \to V$$

such that the following properties hold for all $u, v, w \in V$ and all $a, b \in \mathbf{F}$:

(i) $u + v \in V$;

(ii) $u + v = v + u$;

(iii) $(u + v) + w = u + (v + w)$;

(iv) there exists $0 \in V$ such that $v + 0 = v$;

(v) for every $v \in V$ there exists $-v \in V$ such that $v + (-v) = 0$;

(vi) $av \in V$;

(vii) $a(u + v) = au + av$;

(viii) $(a + b)v = av + bv$;

(ix) $a(bv) = (ab)v$;

(x) $1v = v$.

Definition 0.17.9 (Vector and Point)**.** An element of a vector space V is called a **vector**.
When $V = \mathbf{F}^n$, an element of $\mathbf{F}^n$ may also be called a **point**.

Definition 0.17.10 (Real Vector Space)**.** A **real vector space** is a vector space over $\mathbb{R}$.

Definition 0.17.11 (Complex Vector Space)**.** A **complex vector space** is a vector space over $\mathbb{C}$.

Theorem 0.17.12 (Unique Additive Identity)**.** *A vector space has a unique additive identity.* *Go to proof.*

Theorem 0.17.13 (Unique Additive Inverse)**.** *Every vector in a vector space has a unique additive inverse.* *Go to proof.*

Theorem 0.17.14 ($0v = 0$)**.** *For every $v \in V$,*

$$0v = 0.$$

Go to proof.

Theorem 0.17.15 ($a0 = 0$)**.** *For every $a \in \mathbf{F}$,*

$$a0 = 0.$$

Go to proof.

Theorem 0.17.16 ($(-1)v = -v$)**.** *For every $v \in V$,*

$$(-1)v = -v.$$

Go to proof.

Definition 0.17.17 (Subspace)**.** Let V be a vector space over $\mathbf{F}$. A subset $U \subseteq V$ is called a **subspace** of V if U is itself a vector space over $\mathbf{F}$ with the same addition and scalar multiplication as on V.

Theorem 0.17.18 (Conditions for a Subspace)**.** *Let V be a vector space over $\mathbf{F}$ and let $U \subseteq V$. Then U is a subspace of V if and only if the following three conditions hold:*

(i) $0 \in U$*;*

(ii) for all $u, v \in U$, one has $u + v \in U$;

(iii) for all $a \in \mathbf{F}$ and all $u \in U$, one has $au \in U$.

Go to proof.

Definition 0.17.19 (Sum of Subspaces)**.** Let V be a vector space over $\mathbf{F}$, and let $U_1, \ldots, U_m$ be subspaces of V. The **sum** of $U_1, \ldots, U_m$ is the set

$$U_1 + \cdots + U_m = \{u_1 + \cdots + u_m : u_1 \in U_1, \ldots, u_m \in U_m\}.$$

Theorem 0.17.20 (Sum of Subspaces Is the Smallest Containing Subspace)**.** *Let V be a vector space over $\mathbf{F}$, and let $U_1, \ldots, U_m$ be subspaces of V. Then $U_1 + \cdots + U_m$ is a subspace of V containing each of $U_1, \ldots, U_m$. Moreover, if W is a subspace of V containing each of $U_1, \ldots, U_m$, then*

$$U_1 + \cdots + U_m \subseteq W.$$

Go to proof.

Definition 0.17.21 (Direct Sum)**.** Let V be a vector space over $\mathbf{F}$, and let $U_1, \ldots, U_m$ be subspaces of V such that

$$V = U_1 + \cdots + U_m.$$

Then V is called the **direct sum** of $U_1, \ldots, U_m$ if each vector $v \in V$ can be written in the form

$$v = u_1 + \cdots + u_m, \qquad u_1 \in U_1, \ldots, u_m \in U_m,$$

in exactly one way. In this case, we write

$$V = U_1 \oplus \cdots \oplus U_m.$$

Theorem 0.17.22 (Condition for a Direct Sum)**.** *Let V be a vector space over $\mathbf{F}$, and let $U_1, \ldots, U_m$ be subspaces of V such that*

$$V = U_1 + \cdots + U_m.$$

Then the following are equivalent:

(i) $V = U_1 \oplus \cdots \oplus U_m$*;*

(ii) if $u_1 \in U_1, \ldots, u_m \in U_m$ *and*

$$u_1 + \cdots + u_m = 0,$$

then

$$u_1 = \cdots = u_m = 0.$$

Go to proof.

Theorem 0.17.23 (Direct Sum of Two Subspaces)**.** *Let U and W be subspaces of a vector space V. Then*

$$U + W = U \oplus W$$

if and only if

$$U \cap W = \{0\}.$$

Go to proof.

Proofs

Capstone

Algebra

Lattice Order

0.18 Breadcrumb

Why Lattice Theory Gets Its Own Home

0.18.1 Ordered Sets

Definition 0.18.1 (Partial Order)**.** A relation $\leq$ on a set S is a *partial order* if it satisfies, for all $a, b, c \in S$:

1. **Reflexivity:** $a \leq a$.
2. **Anti-symmetry:** $a \leq b \wedge b \leq a \Longrightarrow a = b$.
3. **Transitivity:** $a \leq b \wedge b \leq c \Longrightarrow a \leq c$.

The pair $(S, \leq)$ is called a *partially ordered set*, or *poset*.

Definition 0.18.2 (Strict Partial Order)**.** A relation $<$ on a set S is a *strict partial order* if it satisfies, for all $a, b, c \in S$:

1. **Irreflexivity:** $\neg(a < a)$.
2. **Asymmetry:** $a < b \Longrightarrow \neg(b < a)$.
3. **Transitivity:** $a < b \wedge b < c \Longrightarrow a < c$.

Definition 0.18.3 (Total Order)**.** A partial order $\leq$ on a set S is a *total order* (or *linear order*) if it additionally satisfies the *law of trichotomy*: for all $a, b \in S$, exactly one of

$$a < b, \qquad a = b, \qquad b < a$$

holds. The pair $(S, \leq)$ is called a *totally ordered set* or a *chain*.

0.18.2 Chains and Antichains

Definition 0.18.4 (Chain)**.** Let $(S, \leq)$ be a partially ordered set. A subset $C \subseteq S$ is a *chain* if every pair of elements in C is comparable:

$$\forall x, y \in C, \quad x \leq y \ \vee \ y \leq x.$$

Equivalently, the restriction of $\leq$ to C is a total order on C.

Definition 0.18.5 (Antichain)**.** Let $(S, \leq)$ be a partially ordered set. A subset $A \subseteq S$ is an *antichain* if no two distinct elements of A are comparable:

$$\forall x, y \in A, \quad x \neq y \Longrightarrow \neg(x \leq y) \wedge \neg(y \leq x).$$

0.18.3 Partial and Total Maps

Definition 0.18.6 (Partial Map)**.** Let A and B be sets. A *partial map* $f : A \rightharpoonup B$ is a relation $f \subseteq A \times B$ such that every element of A has at most one image:

$$\forall x \in A,\ \forall y_1, y_2 \in B, \quad (x, y_1) \in f \ \wedge\ (x, y_2) \in f \implies y_1 = y_2.$$

Definition 0.18.7 (Total Map)**.** A partial map $f : A \rightharpoonup B$ is a *total map* if it is defined on every element of A:

$$\forall x \in A,\ \exists y \in B, \quad (x, y) \in f.$$

A total map is written $f : A \to B$ and is precisely a function in the usual sense.

0.18.4 Maps Between Ordered Sets

Definition 0.18.8 (Order-Preserving Map)**.** Let $(P, \leq_P)$ and $(Q, \leq_Q)$ be partially ordered sets. A map $\varphi : P \to Q$ is *order-preserving* (or *monotone*) if

$$\forall x, y \in P, \quad x \leq_P y \implies \varphi(x) \leq_Q \varphi(y).$$

Definition 0.18.9 (Order-Embedding)**.** Let $(P, \leq_P)$ and $(Q, \leq_Q)$ be partially ordered sets. A map $\varphi : P \to Q$ is an *order-embedding* if

$$\forall x, y \in P, \quad x \leq_P y \iff \varphi(x) \leq_Q \varphi(y).$$

When φ is an order-embedding, one writes $\varphi : P \hookrightarrow Q$.

Definition 0.18.10 (Order-Isomorphism)**.** An order-embedding $\varphi : P \to Q$ that is surjective is an *order-isomorphism*. When an order-isomorphism exists, P and Q are said to be *order-isomorphic*, written $P \cong Q$.

Lemma 0.18.11 (Every Order-Embedding Is Injective)**.** *Let $(P, \leq_P)$ and $(Q, \leq_Q)$ be partially ordered sets, and let $\varphi : P \to Q$ be an order-embedding. Then φ is injective.* *Go to proof.*

0.18.5 The Covering Relation

Definition 0.18.12 (Covering Relation)**.** Let $(P, \leq)$ be a partially ordered set and let $x, y \in P$ with $x < y$. The element y *covers* x, written $x \prec y$, if there is no element of P strictly between them:

$$x \prec y \ :=\ x < y \ \wedge\ \neg\big(\exists z \in P,\ x < z < y\big).$$

Lemma 0.18.13 (Characterizations of Order-Isomorphisms for Finite Posets)**.** *Let $(P, \leq_P)$ and $(Q, \leq_Q)$ be finite partially ordered sets and let $\varphi : P \to Q$ be a bijection. The following are equivalent:*

1. *φ is an order-isomorphism.*
2. *For all $x, y \in P$: $x <_P y \iff \varphi(x) <_Q \varphi(y)$.*
3. *For all $x, y \in P$: $x \prec_P y \iff \varphi(x) \prec_Q \varphi(y)$.*

Go to proof.

Proposition 0.18.14 (Hasse Diagram Characterization for Finite Posets)**.** *Let $(P, \leq_P)$ and $(Q, \leq_Q)$ be finite partially ordered sets. Then P and Q are order-isomorphic if and only if their Hasse diagrams are isomorphic as directed graphs — that is, identical up to relabeling of elements.* *Go to proof.*

0.18.6 Duality

Definition 0.18.15 (Dual Ordered Set)**.** Let $(P, \leq)$ be a partially ordered set. The *dual* (or *opposite*) of P, denoted P^{op}, is the set P equipped with the reversed order:

$$x \leq_{P^{\mathrm{op}}} y \iff y \leq_P x.$$

Proposition 0.18.16 (Duality Principle)**.** *Let Φ be a statement about partially ordered sets, expressed purely in terms of $\leq$, that holds in every partially ordered set. Then the dual statement Φ^{op}, obtained by replacing every occurrence of $\leq$ with $\geq$, also holds in every partially ordered set.* *Go to proof.*

0.18.7 Extremal Elements

Definition 0.18.17 (Bottom Element)**.** Let $(P, \leq)$ be a partially ordered set. An element $\bot \in P$ is a *bottom element* (or *least element*) of P if

$$\forall x \in P, \quad \bot \leq x.$$

If a bottom element exists it is unique; it is denoted $\bot_P$ or simply $\bot$.

Definition 0.18.18 (Top Element)**.** Let $(P, \leq)$ be a partially ordered set. An element $\top \in P$ is a *top element* (or *greatest element*) of P if

$$\forall x \in P, \quad x \leq \top.$$

If a top element exists it is unique; it is denoted $\top_P$ or simply $\top$.

Definition 0.18.19 (Maximal Element)**.** Let $(P, \leq)$ be a partially ordered set and let $Q \subseteq P$. An element $a \in Q$ is *maximal in* Q if no element of Q strictly exceeds it:

$$\forall x \in Q, \quad a \leq x \implies a = x.$$

The set of maximal elements of Q is denoted $\mathrm{Max}(Q)$.

Definition 0.18.20 (Minimal Element)**.** Let $(P, \leq)$ be a partially ordered set and let $Q \subseteq P$. An element $a \in Q$ is *minimal in* Q if no element of Q is strictly below it:

$$\forall x \in Q, \quad x \leq a \implies x = a.$$

Definition 0.18.21 (Greatest Element)**.** Let $(P, \leq)$ be a partially ordered set and let $Q \subseteq P$. An element $m \in Q$ is the *greatest element* (or *maximum*) of Q if it dominates every element of Q:

$$\forall x \in Q, \quad x \leq m.$$

If it exists it is unique; it is denoted $\max Q$.

Definition 0.18.22 (Least Element)**.** Let $(P, \leq)$ be a partially ordered set and let $Q \subseteq P$. An element $m \in Q$ is the *least element* (or *minimum*) of Q if it lies below every element of Q:

$$\forall x \in Q, \quad m \leq x.$$

If it exists it is unique; it is denoted $\min Q$.

Proposition 0.18.23 (Every Nonempty Finite Poset Has a Maximal Element)**.** *Let $(P, \leq)$ be a finite partially ordered set. Then every nonempty subset $Q \subseteq P$ has at least one maximal element. Furthermore, for every $x \in P$ there exists $y \in \mathrm{Max}(P)$ with $x \leq y$.* *Go to proof.*

0.18.8 Lifting and Co-Lifting

Definition 0.18.24 (Lifting)**.** Let $(P, \leq)$ be a partially ordered set. The *lifting* of P is the poset $P_\bot := P \cup \{\bot\}$, where $\bot \notin P$, with the order:

$$\bot \leq x \quad \text{for all } x \in P_\bot, \qquad x \leq_{P_\bot} y \iff x \leq_P y \quad \text{for all } x, y \in P.$$

Thus $P_\bot$ is P with a new bottom element $\bot$ adjoined.

Definition 0.18.25 (Co-Lifting)**.** Let $(P, \leq)$ be a partially ordered set. The *co-lifting* of P is the poset $P^\top := P \cup \{\top\}$, where $\top \notin P$, with the order:

$$x \leq \top \quad \text{for all } x \in P^\top, \qquad x \leq_{P^\top} y \iff x \leq_P y \quad \text{for all } x, y \in P.$$

Thus $P^\top$ is P with a new top element $\top$ adjoined.

0.18.9 Consistency

Definition 0.18.26 (Consistency)**.** Let $(S, \leq)$ be a partially ordered set. Elements $x, y \in S$ are *consistent* (or *compatible*) if they admit a common upper bound:

$$\exists z \in S, \quad x \leq z \ \wedge \ y \leq z.$$

0.19 Ordered Sets

0.19.1 Down-Sets and Up-Sets

Definition 0.19.1 (Down-Set)**.** Let $(P, \leq)$ be a partially ordered set. A subset $Q \subseteq P$ is a *down-set* (or *decreasing set*, or *order ideal*) if it is closed downward:

$$\forall x \in Q,\ \forall y \in P, \quad y \leq x \implies y \in Q.$$

Definition 0.19.2 (Up-Set)**.** Let $(P, \leq)$ be a partially ordered set. A subset $Q \subseteq P$ is an *up-set* (or *increasing set*, or *order filter*) if it is closed upward:

$$\forall x \in Q,\ \forall y \in P, \quad x \leq y \implies y \in Q.$$

0.19.2 Families of Sets

Proposition 0.19.3 (Predicates and Power Sets Are Order-Isomorphic)**.** *The map $\Phi : (\mathcal{P}^*(X), \leq) \to (\mathcal{P}(X), \subseteq)$ is an order-isomorphism. Go to proof.*

Proofs

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